

Statistical Modelling and Inference II

Assignment 2

Solution

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ASSIGNMENT CHECKLIST

- Have you shown all of your working, including probability notation where necessary?
- Have you included all R output and plots to support your answers where necessary?
- Have you included all of your R code?
- Have you made sure that all plots and tables each have a caption?
- Is the submission a single pdf file - correctly orientated, easy to read? If not, penalties apply.
- Assignments emailed instead of submitted by the online submission on BB will not be marked and will receive zero.
- Q1-Q3: may be typed or hand written and scanned in as a pdf
- Q4: pls upload your R code to BB

Q1

This questions may be typed or hand written and scanned in as a pdf.

The aim of this question is to build your experience with power and sample size calculations. You are introduced to calculating the power for a two-sided normal Z test, and applying the sample size calculation to this test.

A study was conducted to compare finger dexterity of music students who study piano to those who study singing. A random sample of 137 students from the piano group had mean dexterity score of 37.25 and standard deviation 4.34. An independent sample of 137 singing students had mean dexterity score of 35.91 and standard deviation of 5.19. You may assume the data is normally distributed. Let μ_1 and σ_1^2 denote the true mean and variance dexterity score of piano students and μ_2 and σ_2^2 denote the true mean and variance dexterity score of singing students.

- (a) Test to see whether there is sufficient evidence in the sample to indicate that piano students have a different mean dexterity score than singing students. You may use a significance level of 0.05 and make the assumption $\sigma_1 \approx 4.34$ and $\sigma_2 \approx 5.19$. [4 marks]
- (b) Calculate the power of the test used in part (a) when $\mu_1 - \mu_2 = 3$. [4 marks]
- (c) Find the sample size needed for the test to achieve a significance level of 0.05 and power of 0.95, when $\mu_1 - \mu_2 = 3$. You may assume equal sample size for both group and that $\sigma_1^2 = \sigma_2^2 = \sigma^2$. **Hint.** Estimate σ^2 with

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}.$$

[2 marks]

[Question total: 10]

Solutions:

(a) Let

- $\bar{Y}_1$ denote the sample mean dexterity score of piano students, and
- $\bar{Y}_2$ denote the sample mean dexterity score of singing students.

Using the approximation $\sigma_1 \approx 4.34$ and $\sigma_2 \approx 5.19$, we can use the two-sample z-test. Observe that

- $E[\bar{Y}_1 - \bar{Y}_2] = \mu_1 - \mu_2$, and
- $\text{var}(\bar{Y}_1 - \bar{Y}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} = \frac{4.34^2 + 5.19^2}{137} = 0.3341$.

We are testing

$$H_0 : \mu_1 - \mu_2 = 0 \quad \text{vs} \quad H_a : \mu_1 - \mu_2 \neq 0.$$

The test statistic is

$$z = \frac{\bar{y}_1 - \bar{y}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \approx 2.318$$

The critical region is given by $|z^*| > z_{0.025} = 1.96$. Hence we reject the null hypothesis since $z > 1.96$ and conclude there is sufficient evidence at the 5% level to suggest that piano students have a different mean dexterity score than singing students. ***(Through the sign of the test statistics, this suggests piano students have a higher mean dexterity score)***

- (b) The critical region from part (a) was $|z| > 1.96$. This is equivalent to $|\bar{y}_1 - \bar{y}_2| > 1.96\sqrt{0.3341} \approx 1.1329$. Now observe that, since $\mu_1 - \mu_2 = 3$, we have $\bar{Y}_1 - \bar{Y}_2 \sim N(3, 0.3341)$. Then the power of this test is

$$\begin{aligned} \text{Power} &= 1 - \beta \\ &= P(\text{reject } H_0 \mid H_0 \text{ false}) \\ &= P(|\bar{Y}_1 - \bar{Y}_2| > 1.1329 \mid \mu_1 - \mu_2 = 3) \\ &= P(\bar{Y}_1 - \bar{Y}_2 < -1.1329 \mid \mu_1 - \mu_2 = 3) + P(\bar{Y}_1 - \bar{Y}_2 > 1.1329 \mid \mu_1 - \mu_2 = 3) \\ &= P\left(\frac{(\bar{Y}_1 - \bar{Y}_2) - 3}{\sqrt{0.3341}} < \frac{-1.1329 - 3}{\sqrt{0.3341}}\right) + P\left(\frac{(\bar{Y}_1 - \bar{Y}_2) - 3}{\sqrt{0.3341}} > \frac{1.1329 - 3}{\sqrt{0.3341}}\right) \\ &= P(Z < -7.15) + 1 - P(Z < -3.23) \\ &= 0.999381. \end{aligned}$$

(c) First estimate σ^2 by

$$\sigma^2 \approx s_p^2 = \frac{(n_1 - 1)\sigma_1^2 + (n_2 - 1)\sigma_2^2}{n_1 + n_2 - 2} = \frac{(136)(4.34^2 + 5.19^2)}{137 + 137 - 2} \approx 22.89.$$

Then, as given in lecture, the sample size for a two-sample one-sided z-test (assuming equal variance) is

$$n = \frac{2\sigma^2(z_{\alpha/2} + z_{\beta})^2}{(\mu_a - \mu_0)^2} = \frac{2(22.89)(1.96 + 1.645)^2}{3^2} = 66.12$$

Alternatively, you may also use the formula for the case of unequal-variance z-test, given by

$$n = \frac{(\sigma_1^2 + \sigma_2^2)(z_{\alpha/2} + z_{\beta})^2}{(\mu_a - \mu_0)^2},$$

which will give the same result. Hence, we need at least 67 samples in each group.

Q2

This questions may be typed or hand written and scanned in as a pdf.

The aim of this question is to give you experience at performing hypothesis tests and confidence intervals for the population variance. You will be given an opportunity to consolidate your learning of this material from Week 5 and 6.

Consider the study from the previous question to compare finger dexterity of music students who study piano to those who study singing. A random sample of 137 students from the piano group had mean dexterity score of 37.25 and standard deviation 4.34. An independent sample of 137 singing students had mean dexterity score of 35.91 and standard deviation of 5.19. You may assume the data is normally distributed. Let σ_1^2 and σ_2^2 denote the true variance of dexterity score of piano students and singing students, respectively.

- (a) State the sampling distribution of $\frac{(n_1 - 1)S_1^2}{\sigma_1^2}$, where n_1 is the number of students in the piano group and S_1^2 denotes the sample variance of the piano group.

[1 mark]

- (b) Calculate a symmetric 95% confidence interval for σ_1^2 .

[4 marks]

- (c) The researcher has questioned whether they can assume equal variance between the two groups. Perform the hypothesis test

$$H_0 : \sigma_1^2 - \sigma_2^2 = 0 \quad \text{vs} \quad H_a : \sigma_1^2 - \sigma_2^2 \neq 0,$$

at the $\alpha = 0.05$ level of significance. Your hypothesis test should include:

- Your test statistic and the distribution of the test statistic under the null hypothesis.
- The observed value of your test statistic.
- The critical region for this hypothesis test.
- Your conclusion in context.

[7 marks]

[Question total: 12]

Solutions:

- (a)

$$\frac{(n_1 - 1)S_1^2}{\sigma_1^2} \sim \chi_{136}^2$$

- (b) The symmetric 95% CI for σ_1^2 is given by

$$\left(\frac{(n_1 - 1)S_1^2}{c_2}, \frac{(n_1 - 1)S_1^2}{c_1} \right)$$

where

$$P(X < c_1) = P(X > c_2) = 0.025$$

when $X \sim \chi_{136}^2$. Using R, we have:

$$\begin{aligned}c_1 &= 105.6093, \\c_2 &= 170.1753.\end{aligned}$$

Using this, we find that:

$$\begin{aligned}\text{CI} &= \left(\frac{(n_1 - 1)S_1^2}{c_2}, \frac{(n_1 - 1)S_1^2}{c_1} \right) \\&= \left(\frac{(137 - 1)4.34^2}{170.1753}, \frac{(137 - 1)4.34^2}{105.6093} \right) \\&= (15.053, 24.2558) .\end{aligned}$$

(c) Consider the hypothesis test

$$H_0 : \sigma_1^2 - \sigma_2^2 = 0 \quad \text{vs} \quad H_a : \sigma_1^2 - \sigma_2^2 \neq 0 ,$$

at the $\alpha = 0.05$ level of significance. The appropriate test statistic is

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} = \frac{S_1^2}{S_2^2} \sim F_{136,136}$$

under the null hypothesis. The observed value of our test statistic is

$$f = 0.6993 .$$

The Critical region for this test is:

$$\begin{aligned}C_{0.05} &= \{f : f > F_{136,136,0.025} \text{ or } f < F_{136,136,0.975}\} \\&= \{f : f > 1.4015 \text{ or } f < 0.7135\} .\end{aligned}$$

Since $f < 0.7135$, there is sufficient evidence to reject the null hypothesis at the 5% level. In context, this says that it is unreasonable to assume equal variance between the piano group and the singing group.

Q3

This questions may be typed or hand written and scanned in as a pdf.

The aim of this question is is to derive a 1-sample hypothesis test for the population variance. You will be guided through constructing a hypothesis test for σ^2 based on normal data.

Let $Y_1, Y_2, \dots, Y_n$ be i.i.d. normally distributed data with true mean μ and true variance σ^2 . Let S^2 denote the sample variance of the data. Consider the null hypothesis

$$H_0 : \sigma^2 = \sigma_0^2 .$$

- (a) Propose an appropriate 2-sided alternative hypothesis. [1 mark]
- (b) Propose an appropriate test statistic under the null hypothesis, including a statement of the appropriate reference distribution. [2 marks]
- (c) Working at the α level of significance, what is the critical region for this hypothesis test? [2 marks]
- (d) Suggest an appropriate p-value based on your test. [1 mark]

- (e) A researcher is interested in the variability of canine toe nail length. They have measured the toe nail length of 164 randomly selected dogs and found a sample standard deviation of 1.5cm. Assuming the data is normally distributed, test the null hypothesis that the true variance of toe nail length in dogs is 2 at the $\alpha = 0.05$ level.

[6 marks]

[Question total: 12]

Solutions:

- (a) An appropriate alternative hypothesis is

$$H_0 : \sigma^2 \neq \sigma_0^2.$$

- (b) An appropriate test statistic would be

$$X = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi_{n-1}^2.$$

- (c) At the α -level of significance, the critical region would be:

$$C_\alpha = \{x : x > \chi_{n-1, \alpha/2}^2 \text{ or } x < \chi_{n-1, 1-\alpha/2}^2\}.$$

- (d) Given the conventions with 2-sided hypothesis with a skewed reference distribution, we take

$$\text{P-value} = 2 \min\{P(X < x), P(X > x)\}$$

where $X \sim \chi_{n-1}^2$ and x is the observed value of our test statistic.

- (e) We test the hypothesis

$$H_0 : \sigma^2 = 2 \quad \text{vs} \quad H_a : \sigma^2 \neq 2.$$

The test statistic is

$$X = \frac{163S^2}{2} \sim \chi_{163}^2.$$

The observed value is

$$x = 183.375.$$

The critical region is

$$C_{0.05} = \{x : x < 129.5425585 \text{ or } x > 200.2426962\}.$$

Since $x \notin C_{0.05}$, there is insufficient evidence to reject the null hypothesis at the 5% level. This means there is not enough evidence to suggest the true variance of dog toe nail length is different from 2cm.

Alternatively, the P-value is given by

$$\text{P-value} = 2 \min P(X < 183.375), P(X > 183.375) \approx 0.2622.$$

Since the P-value > 0.05 , we have insufficient evidence to reject the null hypothesis.

Q4

Please upload an R script file using BB.

The aim of this question is to help develop your R skills. You will be asked to construct a function to calculate the sample size for a given significance level, proportion, and margin of error. You will then use this function to explore the sample size calculations.

A researcher is planning a pilot study to estimate the proportion of people who are regularly taking over-the-counter paracetamol in South Australia.

- (a) Write a function to calculate the sample size required to estimate a population proportion (**p**), with a desired significance level (**alpha**) and margin of error (**delta**) for a two-sided hypothesis test. Your function should follow the format below, and prints the required sample size (as numeric):

```
proportion_ss <- function(alpha, p, delta){  
  ## your code here  
}
```

- (b) Suppose 20% of the population is regularly taking paracetamol, use your function to calculate the required sample size for 5% significance level with $\pm 10\%$ margin of error. **Hint:** remember to always round up to the nearest integer.
- (c) With COVID-19, the researcher is concerned that the proportion of South Australians regularly taking paracetamol would increase. They wanted to know how this would affect the required sample size.
- Use your function to produce a plot of the required sample size (Y-axis) against proportions (X-axis) ranging from 10% to 90%, assuming 5% significance level with $\pm 10\%$ margin of error.
 - What is the required sample size if 60% of the population is regularly taking paracetamol, assuming 5% significance level with $\pm 10\%$ margin of error?
- (d) The researcher wanted to aim for a lower margin of error.
- Use your function to produce a plot of the required sample size (Y-axis) against proportions (X-axis) ranging from 10% to 90%, with separate lines showing the required sample sizes for each of $\pm 5\%$, $\pm 8\%$ and $\pm 10\%$ margin of error. **Hint:** remember to clearly label each line with a figure legend.
 - What is the required sample size if 50% of the population is regularly taking paracetamol, assuming 5% significance level with $\pm 7\%$ margin of error?
- (e) The researcher is granted ethics approval to approach 100 participants only.
- What margin of error can be achieved with this sample size, assuming a population proportion of 50% and 5% significance level?
 - A margin of error of 8% is desired. Assuming 5% significance level, what is the population proportion/s that can be estimated with this level of margin of error?

Your R script file should contain the function `proportion_ss`, the code to produce the required plots.

For full marks you must include commented code to explain the steps in your function.

[20 marks]

[Question total: 20]

Solutions:

[See the practical R code solution and quiz.](#)

[[Assignment total: 54]]